Cross-Domain Skill Composition: An Exploratory Study of Generated Policies

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Abstract

We report an exploratory study of symbolic policies generated under prompt-level composition of biological/HCI exploration and minimalist-design guidance. Within five fixed retrospective cases, thermal-priority policy A2 is the central candidate: it has the highest observed mean among 30 original Thermal winners, but its value depends on the alternative. It beats a seven-node cubic policy on 390 of 600 saved traces; against the strongest selection-ranked neutral winner, the count narrows to 308, with a small mean margin and two fewer nodes. Observer-instrumented local replay exactly reproduces all 2,400 saved policy-seed rewards for A2 and three fixed alternatives. Across all 600 seeds, positive hazard-credit differences are partly offset by penalties. Against the neutral alternative, 272 traces finish ahead despite losing at the first differing action; 36 do the reverse. Replay adds diagnostics, not independent observations. A1 shows how broader comparison weakens an initially favorable judgment. Both highlighted structures occur in neutral conditions, and the original framing contrast remains -0.11 near-reference components per session. The contribution is a reproducible assessment of generated candidates, their alternatives and their observed tradeoffs. Cross-domain causation and practical utility remain unresolved.

Keywords— *Cross-domain skill composition, symbolic policies, exploratory study, candidate evaluation.*

I. Introduction and Related Work

An interesting generated rule should be something a reader can inspect, explain and compare with an alternative. The possibility of combining perspectives from unrelated domains motivates this work. Our exploratory question is which policies generated under that setting merit further examination when their inputs, alternatives and losses are made explicit. We combine biological immune-adaptation and human-computer interaction (HCI) reframing with minimalist industrial-design guidance. The source domains are named by the prompt; the model chooses expressions within a supplied grammar, task facts, mathematical hints H and feedback. Here skill composition means textual guidance in one prompt, not independently executed agents or unconstrained discovery of source domains.

For reading the comparisons, **X** names cross-domain wording and **N** neutral wording. **ROLE** assigns exploration and simplicity distinct functions; **FLAT** combines the guidance. These are prompt conditions, not separate agents. **H** denotes supplied mathematical hints. A **block** contains one session per condition on the same task. Each session has one **original winner**, chosen on selection data before test access. **Utility U** is mean reward divided by the same task's reference mean; U=1 is the simple reference and larger is better. **Nodes** measure expression size, not runtime. A **trace** is one 320-step synthetic environment run; it is not an independent model generation.

A2 leads as a follow-up candidate; A1 shows how a wider comparison changes judgment. This retrospective emphasis retains all five case IDs and original winners. Sections 2-3 examine outputs and alternatives; Section 5 retains the aggregate evidence.

LLM-driven executable search has precedents in FunSearch [1], Evolution of Heuristics [2] and ReEvo [3]. Analogical prompting [10] and cross-domain scientific search [7] motivate source perspectives; CreativeDC [6] combines divergent and convergent instructions. HSEvo [4], Language Model Crossover [5] and MAP-Elites [9] address diversity alongside quality; persona effects are task-dependent [8]. Our contribution is a traceable case study of generated structures and decisions about their value, not a new search or analogy algorithm. An older 414-expression corpus is excluded because its generation requests are unavailable.

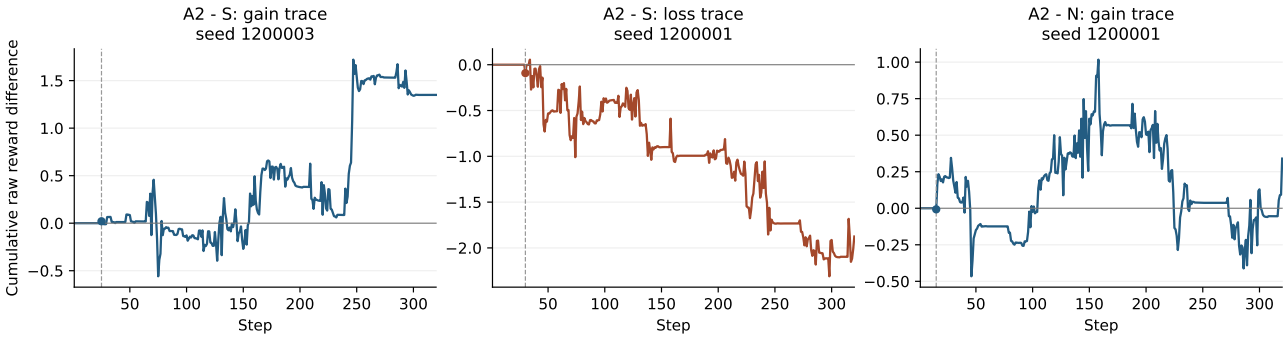


Fig. 1. Cumulative raw reward differences from deterministic replay of existing test seeds. Dashed lines and dots mark the first differing action. Panels use the stated sign-conditioned rule, not the largest gains/losses. The N trace starts with an immediate loss at divergence but ends positive; the mean margin is the final cumulative difference divided by 320. Later paths may contain different states. All six trace pairs are available in the supplement.

II. A2 as a Candidate and A1 under Broader Comparison

The common cross-domain instruction names biological immune-adaptation and HCI reframing for exploration, with minimalist industrial-design guidance for concise expressions. It asks the model to interpret supplied templates; it adds no physical laws or mathematical rules. Table 1 separates those templates from the submitted expressions and the alternatives used to judge them. A1/A2/A3/A5 use X_ROLE; A4 uses X_FLAT. Their presence here records outputs under those prompts, not the origin of the model's internal knowledge.

A. A2: Central Candidate for Further Examination

A2 ranks four synthetic modules for derating using temperature, half the supplied soc*imped interaction and cubic stress. Here soc denotes state of charge and imped impedance. Its highest observed mean among original Thermal winners motivates follow-up, subject to size and loss tradeoffs. On normalized [0,1] inputs, cubing reduces stress except at the endpoints; H supplies a positive-part threshold instead. This is a generated shape within the grammar, not an identified biological/HCI mechanism.

We replay these four existing policies on all 600 original test seeds for 320 steps. An observer records inputs, choices and reward components without changing the original transitions or random draws. All 2,400 policy-seed means match the saved rewards and the uninstrumented simulator exactly; all 600 reference rewards also match. These are additional diagnostics of existing synthetic observations, not new independent tests.

For each alternative we illustrate the smallest base seed with positive A2-minus-alternative mean reward and the smallest with negative mean reward, using tolerance 1e-12. We then inspect the first differing action. This retrospective sign-conditioned rule avoids maximizing the displayed difference but does not make the six pairs representative. Table 3 shows the two selected modules for S; all four inputs and complete trajectories are public.

In the gain trace, the added product moves A2 from S's module 4 to module 3: immediate reward rises from 0.757709 to 0.777924, and the full-trace mean margin is +0.004218. In the loss trace, A2 selects module 4 with higher soc, imped and stress instead of hotter module 1: reward falls from 0.884275 to 0.792774, and the full mean margin is -0.005861. Both examples have zero unnecessary-derating penalty and no floor clipping at that step.

The switching rule is explicit. Let t and x be normalized temperature and stress, p=soc*imped, and B=t+x cubed. For S's selected module b and A2's selected module a, a strict switch requires:

$$0 < B_b - B_a < 0.5(p_a - p_b). \tag{1}$$

The product advantage exceeds S's score disadvantage. In the gain example these quantities are 0.011553 and 0.002476; in the loss example they are 0.014487 and 0.008220. The same rule can therefore accompany either outcome. Relative to C, the added score is 0.4*p; relative to N it is x cubed minus 1.5*max(x-0.5,0). In each case the difference of these added terms between a and b must overcome the alternative's score gap. This describes the selected pair; each policy still takes the maximum over all four modules. Exact ties retain the first module.

The reward uses raw states, whereas policy scores use normalized inputs. Write T, Q, I and X for raw temperature, charge, impedance and stress. For the chosen module, the code computes h=T+0.75*Q*I+0.45*max(X-0.95,0) before acting and D=0.50 only when both X<0.85 and T<0.75 (otherwise zero). The action multiplies its T by 0.45 and X by 0.5. Then P sums 0.4*T+0.28*Q*I across all four post-action modules. The step reward is:

$$r = \max(0.05,\ 0.50 + 0.42h - D - 0.14P). \tag{2}$$

Here h is a designed credit for selecting a high-index module, not measured heat removal. D penalizes an action on a low-state module; P penalizes remaining pack

Case	Supplied input H	Exact generated expression	Comparison alternative	Author judgment
A2	`soc*imped`; `max(stress-0.5,0)`; temperature	`temp+(0.5*(soc* imped)+stress* stress*stress)`	`temp+stress* stress*stress` (X_FLAT; 7 nodes)	Central follow-up candidate: highest observed mean, with extra size and losses
A1	`queue*(0.45+ drift)`; `max(burst-idle,0)`; negative idle	`queue+min(drift, burst)-0.5*idle`	`queue+drift-0.5* idle` (neutral; 7 nodes)	Broader comparison weakens the initial adoption argument; a smaller alternative scores slightly higher
A3	`cost*freq`; `min(freq-0.5, 0.25)`; positive jitter with idle	`idle-0.35* min(freq-0.5, 0.25)+0.2*jitter`	`idle` (reference)	Low-priority diagnostic: supplied cap and tiny reference margin
A4	Same Thermal H as A2	`temp+soc*imped+2* max(stress-0.5,0)`	`temp+soc*imped+ max(stress-0.5,0)` (matched X_ROLE)	Assembly/control example: supplied terms, with higher size and mean
A5	Same Network H as A1	`queue*(0.35+ drift)-0.25*idle`	`queue+drift-0.5* idle` (matched N_ROLE)	Deprioritize: the smaller alternative scores higher

Table 1. Input, exact output, comparison and judgment for all five fixed cases; A2 is presented first and original case IDs are retained. H entries summarize the supplied mathematical hints; the source-domain instruction above, task facts, grammar and session feedback are also inputs. Input-output differences do not identify a biological/HCI mechanism. A1's alternative comes from the full winner set; A2/A4/A5 use matched-block alternatives and A3 uses the reference. These are reasons for further examination or deprioritization, not deployment decisions. Table 2 consolidates A2 and its three nearby alternatives.

Policy	Exact score expression	Nodes	Selection U	Test U
A2	`temp+(0.5*(soc*imped)+stress*stress*stress)`	13	1.011889	1.011170
S	`temp+stress*stress*stress`	7	1.010594	1.009897
N	`temp+0.5*soc*imped+1.5*max(stress-0.5,0)`	15	1.011573	1.011015
C	`temp+stress*stress*stress+0.1*soc*imped`	13	1.011253	1.010659

Table 2. Fixed nearby policies. S is the matched-block seven-node X_FLAT winner; N is the highest selection-mean neutral Thermal winner (N_FLAT block 6); C is the highest selection-mean other 13-node Thermal winner with the literal cubic motif (X_FLAT block 2). These retrospective choices use original winners and selection data, not a coefficient search. U is a ratio of means, not a raw reward difference; nodes measure size, not speed.

state. Let F be the amount added by the 0.05 floor. A2-minus-alternative mean reward decomposes exactly into 0.42 times the mean h difference, minus the mean D difference, minus 0.14 times the mean P difference, plus the mean F difference. These are signed accounting contributions along the observed paths, not causal interventions on individual terms or validated thermal protection.

The saved generation sequence remains visible: round 3 slot 9 contains temp+(soc*imped+stress*stress*stress); round 4 halves the product coefficient after that output and its feedback are supplied. The record does not identify the model's internal rationale.

B. What All 600 Existing Seeds Show

We apply the same accounting and first-action comparison to every saved seed, rather than extrapolating from the six sign-conditioned illustrations. Definitions, signs (tolerance 1e-12), no-divergence handling and tie handling were frozen before this extension; the study and policy choices remain retrospective. Each A2/S, A2/N and A2/C pair differs in action on all 600 seeds. First differences occur at median steps 14, 13 and 16 (ranges 1-90, 1-100 and 1-93). All 1,800 first differences have identical pre-action states and satisfy the strict score-gap identity, with no near-tie or no-divergence cases.

Against S, hazard credit improves on 500/600 seeds, but the pack-penalty contribution is unfavorable on 594/600. The mean positive credit +0.003572 is reduced by action and pack penalties to +0.001635. Against N the net margin is only +0.000198; favorable hazard credit +0.000750 is largely offset by penalties. Against C the same pattern leaves +0.000656. Thus A2 earns more of one designed reward component while accepting other costs, rather than improving every component.

Figure 2 separates the immediate margin at the first different action from the mean over the complete path. Their signs disagree on 295/600 S comparisons, 308/600 N comparisons and 301/600 C comparisons. Against N, 528 first

Trace / step	Module (chosen by)	temp	soc	imped	stress	S score	A2 score
Gain / 25	3 (A2)	0.456673	0.576453	0.224144	0.317894	0.488798	0.553402
Gain / 25	4 (S)	0.461809	0.571182	0.185761	0.308865	0.491274	0.544326
Loss / 30	1 (S)	0.540988	0.593125	0.262826	0.348180	0.583197	0.661141
Loss / 30	4 (A2)	0.462533	0.613333	0.301407	0.482664	0.574977	0.667408

Table 3. Recorded normalized inputs and scores at the first action divergence from S. Gain is base seed 1200003, step 25; loss is seed 1200001, step 30. Steps and modules are one-based. Both policies have identical pre-action states at these points; values are rounded from full precision. The selected module has the highest score among all four modules. Scores are not rewards, and these are synthetic replay states, not physical measurements.

A2 minus	Hazard credit	Action penalty	Pack penalty	Floor	Total
S	+0.003572	-0.000419	-0.001517	0.000000	+0.001635
N	+0.000750	-0.000471	-0.000080	0.000000	+0.000198
C	+0.002263	-0.000388	-0.001219	0.000000	+0.000656

Table 4. Mean signed reward contributions over all 600 original seeds and all 320 steps, in raw reward units. Positive favors A2; a negative penalty contribution means A2 incurred more of that penalty. The four terms sum to the total before rounding, with maximum numerical residual 2.79e-15. The reward floor contributes zero throughout. Values are accounting under the fixed synthetic reward, not percentages or physical benefits.

differences lose immediately: 272 finish positive and 256 negative. Of the 72 initially positive differences, 36 finish positive and 36 negative. A favorable first choice is therefore an inadequate explanation of the observed full-path margin in this sample. After divergence the trajectories can have different states; these counts do not isolate the causal effect of that choice.

The supplement retains all 1,800 per-seed component rows, first-difference inputs/scores, component-sign counts and summaries conditional on final gains/losses. On average, all three non-floor contributions are unfavorable within each comparator's final-loss subset. Conditioning describes those outcomes; it does not predict losses or supply an adoption threshold.

C. *A1: Broader Comparison Changes the Judgment*

A1 ranks synthetic queues by adding the smaller of drift and burst to queue, then subtracting half idle. Both signals must be high for a large added priority. This exact minimum coupling is absent from H.

The same-block comparison initially favors A1: nine nodes and test U=1.003905 versus 13 nodes and U=1.000870 for N_ROLE. Expanding to all original winners reveals the seven-node neutral alternative in Table 1. Section 3 shows why the larger comparison set weakens the adoption judgment even though the minimum rule is easy to explain.

The saved sequence records a coefficient change: round 2 slot 6 contains `queue+min(drift, burst)-0.25*idle`; round 3 changes -0.25 to -0.5 after that output and its feedback are supplied. The sequence establishes the visible revision, not its internal rationale.

D. *A5: Above Reference Is an Incomplete Adoption Argument*

A5 changes a supplied queue template's coefficients and omits burst. Its test U=1.002104 exceeds the simple reference, but the matched neutral expression in Table 1 reaches 1.003917 with seven nodes/depth 2 versus nine/depth 3. This comparison motivates deprioritization rather than labeling A5 a failed generation. Paired outcomes are retained in Section 3.2.

A5 was fixed by the low-selection rule, not chosen for a poor test outcome. The preceding round has `queue*(0.4+drift)-0.3*idle`; A5 revises the coefficients after feedback.

E. *A3 and A4: Supplied Structures Remain Visible*

A3 adjusts idle with an upper-capped frequency term and positive jitter, both supplied in H. For freq below 0.5, subtracting the negative min term adds a bonus; at freq at least 0.75 the contribution is fixed at -0.0875. Earlier rounds use cap weights 1 and 0.6; A3 uses 0.35. Test U=1.000466 and depth 4, above the soft target of 3, make it a low-priority diagnostic rather than evidence of cloud-cost savings.

A4 combines the supplied temperature, soc*imped and threshold terms, doubling the threshold weight. The preceding round contains both the doubled threshold without the product and the product with weight one. Test U=1.009928 exceeds the matched X_ROLE winner's 1.008923, with 13 nodes/depth 4 versus 11/depth 3. It is an X_FLAT assembly/control example, not a newly established domain connection.

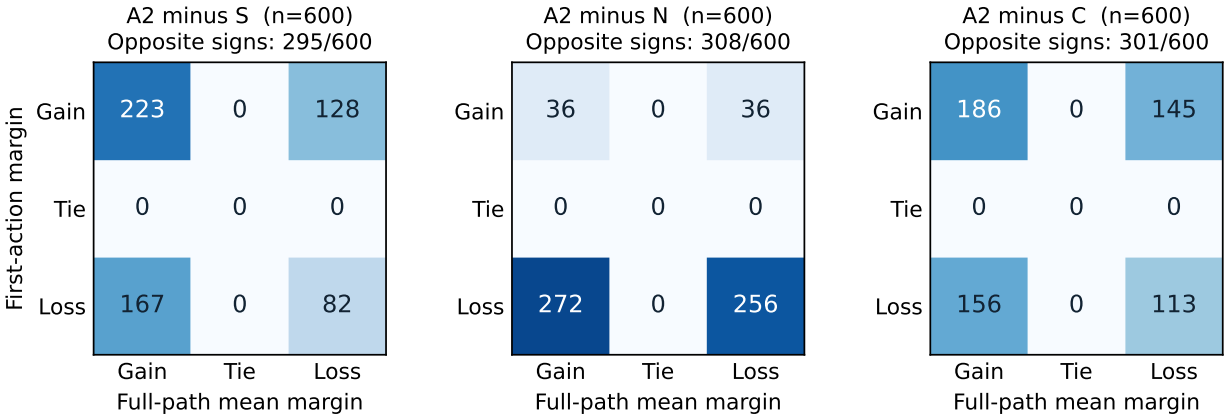


Fig. 2. Counts for the sign of A2's immediate reward margin at the first different action (rows) versus its full-trace mean margin (columns), separately for each alternative; each panel totals 600. Zero rows/columns are displayed, not omitted. Opposite-sign cells reveal how single-step and full-path judgments differ. These are existing seeded traces of fixed policies, not independent model generations.

III. Comparison with All Original Winners

A. *Position among Original Winners and All Submissions*

We rank fixed cases among 30 original winners per task, separately on selection and test, without pooling task utilities. Duplicate outcomes remain: Network and Thermal have 28 and 26 distinct strings. Rank is one plus the number of strictly higher utilities (tolerance 1e-12); this is descriptive, not a generation-effect estimate.

An alternative size-reward dominates a policy if it uses no more nodes and has no lower utility, with one strict difference. We assess each split separately. This rule has no practical-effect threshold; tiny differences can change the status.

A2 ranks 1/30 on both splits and is not size-reward dominated on either; A1 ranks 5/30 and is dominated on both. Among all submitted occurrences, A2 ranks 1/360 on selection and 1/306 available tests; A1 ranks 7/360 and 7/327. The 54 Thermal and 33 Network submissions without tests are not failures. Available-test ranks are not ranks over all 360 submissions.

A2 is not dominated: its test U=1.011170 exceeds the simpler cubic alternative's 1.009897, with 13 rather than seven nodes. A1's two dominators are instances of the same seven-node expression, from Network N_ROLE block 5 and N_FLAT block 6. Its test U=1.003917 is only about 0.000012 above A1; the selection advantage is approximately 4.38e-8. Expression size, rather than a demonstrated practically important score increment, drives the preference. We describe the first instance by session ID; both have identical formulas and stored traces.

Test coverage follows the original evaluation procedure, not random sampling. Public files retain all 90 winner ranks/dominators. The 360-submission view supplies ranks only; per-seed comparisons concern original winners.

B. *Average Gains and Observed Downside*

Table 2 fixes the three nearby alternatives. A2 is two nodes smaller than N with a slightly higher mean, but only 308 wins versus 292 losses; C shares its cubic-product form and differs in weighting. These comparisons qualify candidate value without establishing a unique domain connection.

For each case, we subtract each same-task alternative's raw reward on the identical saved seed. This gives 30 comparisons including self, plus the reference, or 62 comparisons across A1/A2. The within-block contrast remains identifiable, while comparisons to other blocks describe fixed policies rather than paired generation interventions. Joins require 600 matching seeds and equal references.

A win has a difference above 1e-12; a tie is within that tolerance. Table 5 shows the mean difference, the mean of the 30 lowest differences (observed bottom 5%) and the largest observed loss. These are task-specific raw-reward quantities, not percentages. They describe this set of traces, not a bound on future risk or 600 independent model generations.

The earlier comparisons for the other fixed cases are retained: A3 versus reference has 359 wins, no ties and 241 losses, with mean raw margin +0.000412; A4 has 599/0/1 and +0.012758. A5 versus its same-block N_ROLE alternative has 270/2/328 and -0.001764. Their complete matched-condition records remain in the v1.4 analysis.

Gain/loss contributions explain the averages. For A2 versus the simpler cubic rule, +0.002580 gain contribution minus 0.000945 loss contribution gives +0.001635. Against the same-block N_ROLE winner, A1's +0.003617 minus 0.000664 leaves +0.002953; against the smaller cross-block rule, +0.00000355 minus 0.00001530 gives -0.00001176. Contributions divide all positive and absolute negative differences by 600, including differences within the tie tolerance; the supplement also gives conditional averages over only the corresponding signed cases.

Case	Comparator	W/T/L	Mean difference	Lowest-30 mean	Largest loss
A2	Reference	599/0/1	+0.014354	+0.005077	0.000154
A2	S: X_FLAT B5, 7 nodes	390/0/210	+0.001635	-0.006591	0.010608
A2	N: N_FLAT B6, 15 nodes	308/0/292	+0.000198	-0.006281	0.008701
A2	C: X_FLAT B2, 13 nodes	342/0/258	+0.000656	-0.006834	0.012074
A1	Reference	514/0/86	+0.003800	-0.003681	0.007333
A1	N_ROLE B4: same block	402/0/198	+0.002953	-0.004802	0.009958
A1	N_ROLE B5: simpler	9/569/22	-0.000012	-0.000306	0.001276

Table 5. Gain and downside on saved common-seed traces. B denotes block. A negative lowest-30 mean indicates losses in that tail; largest loss is reported as a positive magnitude. The A1 seven-node comparator is identified by the selection-data dominance relation, not by searching test outcomes. The full supplement retains all 62 comparisons, medians, fifth percentiles and conditional gains/losses. S/N/C do not add comparisons to that set; their new step records come from local replay.

These results qualify A2's candidate status: its positive mean coexists with losses against a smaller policy. A1 shows how expanding the comparator set can change an initially favorable adoption judgment. Equal total rewards do not establish identical actions. No acceptable loss or minimum useful gain was specified, so neither tradeoff is labeled practically acceptable. Whole-formula reward differences do not isolate the effect of min or the cubic term.

C. *Interesting Structures Are Not Exclusive to Cross-Domain Roles*

We also audit two explicitly defined motifs throughout the complete saved catalogue. After removing whitespace, we search for the exact substrings `min(drift,burst)` in Network and `stress*stress*stress` in Thermal. This is a narrow, reproducible string census, not semantic clustering or an exhaustive search for algebraically equivalent rules. Counts are retrospective and were chosen because of A1/A2.

The minimum motif appears in four X_ROLE submissions from one session, one N_ROLE submission from one session and one X_FLAT submission from one session. It is absent under this literal rule in N_FLAT and DOMAIN_V2. The cubic motif appears in seven X_FLAT submissions from five sessions, three N_FLAT submissions from one session and two X_ROLE submissions from one session; it is absent under the rule in N_ROLE and DOMAIN_V2. Each task/condition has six sessions and 72 submissions. Repeated submissions from a session are not independent discoveries.

These records sharpen the main distinction: a structure can be absent from H, interpretable and promising in a selected comparison without being specific to the cross-domain package. The minimum and cubic motifs are not evidence that biological, HCI or design knowledge caused the corresponding terms. Their neutral-condition occurrence is part of the finding, not an example to omit.

D. *Complete Collection and Retrospective Selection*

All 1,080 submissions pass the recorded validity gate, with 822 distinct within-domain exact strings (spacing/algebraic variants remain separate), 916 component representative instances and 90 original winners. The catalogue retains every occurrence with complete prompt/response provenance, syntax, selection status, component and scores. Tests exist for 944 submissions; 136 remain untested, not failed.

The offline catalogue supports expression search and provenance/score filters. Syntactic tags locate forms, not semantic novelty or cognitive processes; case cards retain the exact H beside each output.

We fix three core cases: the original X_ROLE session winner with highest selection utility in each domain. Ties use nodes, depth, session ID and slot ID in ascending order. One X_FLAT complement is selected from original winners with a tag combination absent from the core cases, using Network-Thermal-Cloud domain order and the same ranking within the first eligible domain. One low-selection X_ROLE example uses that domain order, minimum selection utility, and the same tie rules after excluding already used domain/expression pairs. A missing eligible complement would be omitted.

The selection function receives only selection/provenance/syntax fields. It writes case IDs and a rule hash before attaching test values, without replacing cases after attachment. Nevertheless, the rule was devised after pilot results had been inspected: these are **post-hoc descriptive cases, not five prevalidated successes or the global best five outputs**. Matched comparison rows are the original winners of all five conditions in the same domain/block. Full inclusion/exclusion logs are public.

IV. Inputs, Generation, and Evaluation

A. *Source Perspectives and Target Tasks*

Biological/HCI and design language supplies guidance; Network, Thermal and Cloud receive generated rules. Network ranks four synthetic queues, Thermal ranks four synthetic modules for derating, and Cloud ranks six synthetic slots for reclamation. In each task, the expression assigns scores to options and the highest score determines the se-

lected option. Features are normalized to [0,1] by clipping negative raw values at zero and applying x/(1+x). Ties retain the first option.

Expressions use four task variables, constants, arithmetic, min, max, and abs. All conditions share a hard gate of 25 AST nodes and implemented depth 5, with soft targets of 14 nodes and depth 3. Recorded depth flattens additive/subtractive and multiplication chains; it is not ordinary binary-tree height. Each environment trace has 320 steps, with handcrafted transitions and rewards; task names do not imply validated physical or production systems.

The four factorial conditions share the hints in Table 7. The cross-domain instruction asks the model to interpret them through biological immune-adaptation and HCI reframing analogies, using minimalist industrial-design inspiration for concise expressions. ROLE assigns exploration and simplicity distinct functions within the same response; FLAT presents both together. These are prompt instructions, not observed internal stages. DOMAIN_V2 omits H and changes the guidance procedure, so it is descriptive rather than a single-module ablation.

Condition	Framing	Organization	H
N_FLAT	Neutral	Combined	Yes
N_ROLE	Neutral	Roles	Yes
X_FLAT	Cross-domain	Combined	Yes
X_ROLE	Cross-domain	Roles	Yes
DOMAIN	Domain	Base	No

Table 6. Conditions. Every factorial arm includes exploration and simplicity guidance. Roles are textual within one response. H denotes shared mathematical hints. DOMAIN abbreviates DOMAIN_V2.

Task	Shared mathematical hints H
Network	`queue*(0.45+drift)`; `max(burst-idle,0)`; negative idle term
Thermal	`soc*imped`; `max(stress-0.5,0)`; temperature term
Cloud	`cost*freq`; `min(freq-0.5,0.25)`; positive jitter with idle

Table 7. H is supplied to all four factorial conditions within a task. The Cloud min term is an upper cap, not a lower clamp. Exact complete prompts and their verbatim differences are public.

B. Recorded Generation and Evaluation

The pilot ran on 24 October 2026 (KST), requesting gpt-6.1-sol with low reasoning effort through Codex CLI 0.162.1. Three domains, six paired blocks per domain, and five conditions give 90 separately initialized sessions. Randomized condition order within domain/block and four rounds of three expressions yield 360 responses and 1,080 slots. Each session receives only its own prior answers and feedback. The required response is an expression array without explanatory commentary, so the record does not reveal the model's internal rationale.

The experimental message roles were serialized into one CLI input rather than sent as native API roles. Actual prompt length, semantics, and token use were not exactly matched; the requested 2,048-token output cap was not enforced. Thus the contrasts concern the recorded instruction packages, not perfectly resource-matched interventions or verified server snapshots.

Feedback, selection and test use disjoint base seeds: 48 (1000001-1000048), 96 (1100001-1100096), and 600 (1200001-1200600), with fixed Thermal/Cloud offsets. Selection starts after generation, and selected policies and representatives are frozen before test access. Independent generation comparisons use sessions; each session's 12 submissions are dependent.

The reference expressions are queue+0.45*drift+0.15*burst-0.15*idle for Network, temp+0.4*soc+0.35*imped+0.25*stress for Thermal, and idle for Cloud. Historical labels Adaptive-PI, PID-lite, and Fixed-TTL denote simplified synthetic formulas, not full implementations of those methods. Utility U is the policy's mean trace reward divided by the reference's mean reward on the same domain/split; the reference has U=1. This ratio is not an average of per-trace ratios. References are comparison points, not optima or tuned published competitors.

Behavioral groups use the saved graph on a fixed four-value-per-coordinate grid and 500 diagnostic option panels. An edge requires Spearman score correlation above 0.98 and top-action agreement of at least 95 percent. Connected components may contain transitive chains; they are finite-probe groups within a session, not globally unique ideas. Each representative and the overall session winner are chosen by selection mean reward, with expression-hash and slot tie breaks, then evaluated on test. The original primary outcome counts representatives with U at least 0.99. This near-reference criterion permits below-reference policies and is not a practical improvement threshold. Strict exceedance uses U greater than 1+1e-9. The analysis was locally frozen before test access, not publicly preregistered.

V. Aggregate Evidence and Boundaries

Selected examples do not replace the original experiment. Equal-domain averaging of six within-block contrasts gives -0.1111 near-reference components per session for X_ROLE minus N_ROLE. The Network, Thermal and Cloud means are -0.6667, +0.1667 and +0.1667, with paired SDs 2.9439, 2.2286 and 4.4907. Figure 3 retains all 18 differences. Six paired blocks per domain are the generation comparisons; test traces do not enlarge that sample.

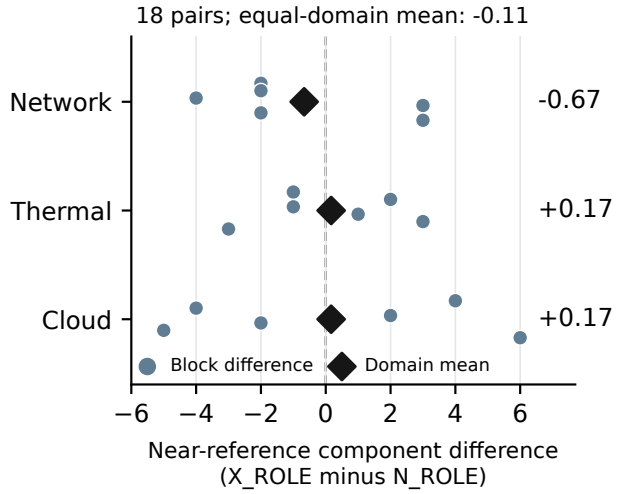


Fig. 3. Original X_ROLE minus N_ROLE near-reference counts by block and domain. Selected artifact examples must be read alongside this mixed aggregate result.

Outcome	Framing	Role	Interaction
Primary count	-0.1111	-0.5556	-1.2222
Strict count	+0.3333	+0.6667	+0.3889
All-pairs count	-0.0556	-0.4444	-1.1111
Policy utility	+0.000998	-0.000144	+0.000062

Table 8. Preserved aggregate comparisons. Interaction is (X_ROLE-X_FLAT)-(N_ROLE-N_FLAT). Counts and utility answer different questions. The utility contrast is not a relative percentage gain. Framing is X_ROLE minus N_ROLE; Role is X_ROLE minus X_FLAT.

The v1.2 retrospective profiles also remain relevant. Relative to N_ROLE, X_ROLE has 0.5370 more mean nodes and 0.3889 fewer pre-threshold components per session; relative to X_FLAT, the differences are +0.8148 and -0.4444. The node difference is mainly Thermal (+1.5556), with Network -0.1667 and Cloud +0.2222. All 12 submissions contribute to session means, including repeats. These profiles do not support an aggregate simplicity-diversity advantage and do not identify a compression process. Full tables and figures remain in the supplement.

Cloud further separates usable-looking artifacts from broad performance claims. Of 304 representative instances, 178 are below 0.99, 118 lie from 0.99 through 1, and eight strictly exceed 1+1e-9, between 1.000012 and 1.000541. Thus 126 meet the near-reference threshold, but only eight exceed the reference; 25 of 30 originally selected Cloud policies are below it. No representatives cross either threshold between saved selection and test splits. This does not rule out selection bias or explain the source of the generated rules.

The common limits apply to all case interpretations. The study uses one requested model/effort, one wording per condition, six paired blocks per task, shared H overlapping reward features, handcrafted transitions/rewards and simple references. Actual token budgets were not matched. Every factorial arm already contains both guidance functions; there are no exploration-only, design-only or bare-length controls. The case rule is retrospective and not representative sampling. Test ranks and observed downside are descriptive; no confidence intervals, equivalence decisions, user ratings or deployment evidence are added.

The catalogue and replay support inspection of outputs and follow-up priorities. They do not reveal which source-domain knowledge was used or show that cross-domain wording caused the structures. Neutral-condition occurrences and the original mixed aggregate result constrain that claim. The supplied soc*imped hint also overlaps the handcrafted reward, which uses the raw-variable product, temperature and a stress threshold. Replay explains choices under that design; it does not remove this overlap. Separating domain-guidance effects and validating practical benefit require evidence beyond these comparisons.

VI. Conclusion and Reproduction

This exploratory study examines generated policies under prompt-level cross-domain skill composition. A2 is the central candidate for further examination within the saved synthetic setting: its observed mean is favorable, while its extra expression size and losses against a smaller policy leave practical adoption open. Across all 600 existing seeds, hazard-credit gains are partly offset by penalties, and the first-action and full-path signs disagree in roughly half of comparisons. These findings explain the candidate's tradeoffs under the fixed synthetic reward without establishing practical utility. A1 shows how expanding beyond a favorable same-block comparison weakens its adoption argument.

The contribution is a traceable account of candidate assessment, linking supplied inputs, exact outputs, alternatives and author judgments. The five retrospective cases, neutral-condition motifs and original mixed aggregate outcome remain visible. These records do not establish that cross-domain knowledge caused the structures or that the policies are useful in deployment; the exploratory scope is part of the conclusion.

The public repository provides the v1.9.0 paper, source, all-seed diagnostics, replay code and Korean guide: `https://github.com/7065616b/symbolic-policy-framing-pilot`. Earlier analyses, catalogues, case cards, exact prompts and evidence archives remain unchanged. The replay reproduces existing seeded environments locally without model calls; identical model regeneration is not claimed because server snapshots and sampling seeds are unavailable. No new policies, seeds or model calls are added. Replayed step records are additional diagnostics, not independent observations; the original data remain unchanged.

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SUMMARY OF THIS PAPER

A. Problem Setup

This exploratory study examines symbolic policies generated under prompt-level composition of biological/HCI exploration and minimalist-design guidance. The prompt names the source perspectives and also supplies task facts, mathematical hints, grammar and feedback. The question is which saved outputs merit further examination, not whether a causal cross-domain benefit has been established.

B. Novelty

The contribution is a traceable case study of candidate assessment. A2 leads as the central candidate with a favorable observed mean and unresolved tradeoffs. A1 illustrates how a wider alternative set can change an initially favorable judgment. Input-output tables and exact local replay explain these fixed retrospective cases without claiming a new algorithm, mathematical invention or module synergy.

C. Algorithms

The original 90 sessions yield 1,080 submissions with separate feedback, selection and test data. All five case IDs and original winners remain fixed; the presentation order does not reselect policies on test results. A1/A2 comparisons cover all 30 original winners in their own task and 62 common-seed comparisons. Secondary ranks disclose missing test values among all 360 task submissions. Four existing policies are replayed on the original 600 seeds, with exact agreement for all 2,400 saved rewards. All 1,800 comparator-seed pairs receive reward-component and first-divergence diagnostics; six sign-conditioned pairs illustrate complete trajectories. Neither increases the independent sample.

D. Experiments

A2 ranks first among 30 Thermal winners on both saved splits. Against the seven-node cubic alternative, its 13-node expression has mean raw margin +0.001635, 390 wins and 210 losses, with largest observed loss 0.010608. Against the strongest selection-ranked neutral winner, A2 has 308 wins and 292 losses, with mean margin +0.000198 and 13 versus 15 nodes. Against N, 272/600 traces finish positive after an initially negative margin, while 36 do the reverse; the mean hazard-credit gain is largely offset by penalties. A1 ranks fifth among 30 Network winners; two instances of a smaller neutral rule have slightly higher means, and the paired results against that rule are 9 wins, 569 ties and 22 losses. A2 ranks first of 360 submissions on selection and first of 306 tested occurrences; A1 ranks seventh of 360 and seventh of 327. The original X_ROLE minus N_ROLE primary contrast remains -0.1111 near-reference components per session, with strict exceedance +0.3333. Neutral-condition motifs, shared hints, one requested model, six blocks per task, synthetic environments and retrospective emphasis limit interpretation. Causal cross-domain effects and practical utility remain unresolved.